

SOP-NDT-3300: Radiographic Testing and Radiation Safety

Document owner: Quality Assurance Department **Effective date:** 2024-05-10 **Review cycle:** 24 months

Section 1. Purpose and Scope

Governs use of radiography (gamma or X-ray) for non-destructive testing of welds and castings, and the radiation safety controls required.

Section 2. Radiation Safety Officer Authority

Clause 2.1 — All radiographic work on site is conducted under the authority of the Radiation Safety Officer (RSO); no radiography shall begin without RSO-issued work authorization for that specific shot location and time.

Section 3. Boundary and Posting

Clause 3.1 — A radiation exclusion boundary is established and surveyed by the radiographer before each exposure, calculated to keep dose outside the boundary below the regulatory public dose limit for the exposure duration; the boundary shall be physically barricaded and posted, not just verbally warned.

Clause 3.2 — Radiographers shall survey the boundary with a calibrated survey meter during the exposure, not rely solely on pre-calculated distances.

Section 4. Dosimetry

Clause 4.1 — Radiographers shall wear a dosimeter and an audible dose-rate alarm at all times during source handling and exposure; a radiographer whose alarm activates outside expected parameters shall stop work immediately and notify the RSO.

Section 5. Source Accountability

Clause 5.1 — Sealed source inventory is verified and logged before and after each job; a source that cannot be immediately accounted for triggers the site's lost-source emergency procedure without delay.

Section 6. Records

Clause 6.1 — Exposure logs, dosimetry records, and source inventory logs are filed to the Radiation Safety File and retained per regulatory requirement (typically the life of the license plus 5 years).